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MINOR PROJECT REPORT

ON

EPILEPSY LESION LOCALIZATION BASED ON BRAIN FUNCTION NETWORK

(A Deep Sequential Learning and Topological Data Analysis Framework)

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DECLARATION

We hereby declare that this minor project report entitled "Epilepsy Lesion Localization Based on Brain Function Network" is our own original work completed under the guidance of Dr. Rekh Ram Janghel, Associate Professor, Department of Information Technology, National Institute of Technology, Raipur. This report contains no material previously published or written by another author except where due academic reference and citation is made in the text. This work has not been submitted for the award of any other degree or diploma.

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ABSTRACT

Drug-resistant epilepsy (DRE) affects approximately 20% to 30% of the worldwide epileptic population, rendering pharmacotherapy ineffective and establishing surgical resection or thermal ablation of the epileptogenic zone (EZ) as the primary curative intervention. Achieving postoperative seizure freedom critically hinges upon the accurate spatial localization of the Seizure Onset Zone (SOZ). Clinically, this requires invasively recording electrophysiological telemetry via multi-lead Stereotactic Electroencephalography (SEEG) depth electrodes. However, manual visual inspection of continuous multi-channel SEEG data across hundreds of electrode contacts is exceptionally labor-intensive, operator-subjective, and susceptible to substantial inter-rater variability. In this minor project, we investigate and advance automated electrophysiological lesion localization by synthesizing dynamic functional brain networks, topological data analysis (TDA), and deep recurrent sequence modeling.

In the mid-semester phase, we replicated the state-of-the-art computational pipeline proposed by Fang et al. (Frontiers in Human Neuroscience, 2024) across 22 drug-resistant epilepsy patients (>2,500 intracranial electrode contacts) from the public Penn HUP-iEEG surgical cohort (OpenNeuro ds004100). The baseline architecture utilizes a dual-branch feature extraction scheme: Branch 1 computes the Weighted Phase Lag Index (WPLI) in the gamma band (30–60 Hz), yielding a 50% proportionally thresholded functional brain network to extract the Epileptic Network Connectivity Strength (ENCS = node degree); Branch 2 applies Takens delay coordinate embedding (delay tau=28, embedding dimension m=4) and Vietoris-Rips simplicial filtration to compute multi-dimensional persistent homology (H0, H1, H2) and Shannon Persistence Entropy (PE). Classification using a static Support Vector Machine (SVM, RBF kernel, C=0.5) achieved 59.69% accuracy using ENCS alone, and 72.21% accuracy when fusing ENCS with Persistence Entropy (+12.52% boost). Crucially, our mid-semester investigations revealed a fundamental architectural limitation: static classifiers treat each 30-second sliding window as an isolated, memoryless snapshot, failing to model the evolving temporal propagation and spatial recruitment of ictal discharges.

To overcome this barrier, in this end-semester completion, we propose and engineer a novel temporal deep learning framework based on a Bidirectional Gated Recurrent Unit (Bi-GRU). The Bi-GRU sequence model explicitly captures the multi-scale temporal trajectory of functional connectivity and topological persistence across consecutive micro-windows. Combined with a newly introduced Seizure-Centric Event-Aligned windowing strategy (10-second micro-windows with 5-second shifts centered on electrographic onset) and dynamic focal cross-entropy loss for extreme clinical class imbalance mitigation, our proposed architecture boosts cohort-wide classification accuracy to 98.66%, precision to 98.09%, recall (sensitivity) to 99.36%, F1-score to 98.71%, and ROC-AUC to 98.91% across all 22 subjects evaluated under rigorous 5-fold stratified cross-validation. Electrode-level spatial mapping confirms near-zero false-negative channel classifications, ensuring complete coverage of surgical resection targets while preserving eloquent cortex. This work demonstrates that recurrent temporal modeling of functional and topological manifolds provides a clinically viable, objective decision-support paradigm for presurgical epilepsy evaluation.

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CHAPTER 1: INTRODUCTION

1.1 Clinical Background of Epilepsy and Drug-Resistant Epilepsy (DRE)

Epilepsy is one of the most prevalent and debilitating chronic neurological disorders worldwide, affecting an estimated 50 million individuals across all age demographics [1]. Characterized by recurrent, unprovoked seizures resulting from paroxysmal, excessive, and hypersynchronous electrical discharges of cerebral neurons, epilepsy severely compromises patient autonomy, cognitive capability, and quality of life [2]. The sudden, uncontrolled bursts of hypersynchronous neuronal firing—frequently characterized as an 'electrical storm' within the brain—disrupt physiological communications, manifesting in loss of consciousness, focal motor convulsions, sensory distortions, and, in severe cases, status epilepticus or sudden unexpected death in epilepsy (SUDEP) [3].

While pharmacotherapy utilizing Anti-Seizure Medications (ASMs) remains the frontline clinical treatment, approximately 20% to 30% of all epilepsy patients exhibit pharmacoresistance, a condition clinically classified as Drug-Resistant Epilepsy (DRE) or medically refractory epilepsy [4]. For these individuals, multiple trials of tolerated and appropriately chosen ASM regimens fail to achieve sustained seizure freedom. In pharmacoresistant focal epilepsy, resective neurosurgery or targeted laser interstitial thermal therapy (LITT) constitutes the primary curative intervention [5]. The ultimate objective of epilepsy surgery is the complete surgical excision or ablation of the Epileptogenic Zone (EZ)—defined as the minimum amount of cortical tissue that must be resected or disconnected to render the patient completely seizure-free (Engel Class I outcome) without inducing new permanent neurological deficits [6].

Clinically, the primary diagnostic proxy for delineating the EZ is the identification of the Seizure Onset Zone (SOZ), which denotes the anatomically defined brain region that initiates clinical or electrographic seizure activity [1]. Accurate spatial localization of the SOZ is the single most critical factor determining postoperative seizure outcomes. Under-resection of the SOZ inevitably leads to persistent seizure recurrence, whereas over-resection unnecessarily destroys adjacent eloquent cortex, causing catastrophic irreversible impairments in speech, memory, motor function, or visual processing [7].

1.2 Stereotactic EEG (SEEG) Modality & Presurgical Evaluation

During Phase I presurgical evaluations, non-invasive neuroimaging modalities—such as scalp Electroencephalography (EEG), high-resolution 3T Magnetic Resonance Imaging (MRI), Positron Emission Tomography (PET), and Single-Photon Emission Computed Tomography (SPECT)—are deployed to formulate a concordant clinicopathological hypothesis [8]. However, in a significant proportion of patients with non-lesional (MRI-negative) epilepsy or deep-seated focal cortical dysplasias (FCD), non-invasive modalities are insufficient. Scalp EEG suffers from profound spatial blurring, high skull impedance, low signal-to-noise ratio (SNR), and severe volume conduction artifacts, making it fundamentally incapable of reliably recording electrical activity originating in deep mesial temporal structures, insular cortex, or interhemispheric sulci [9].

Consequently, patients undergo invasive Phase II presurgical monitoring via Stereotactic Electroencephalography (SEEG) [10]. SEEG is an invasive clinical monitoring technique wherein multiple multi-lead depth electrodes are surgically implanted stereotactically into specific brain regions under robotic guidance. Each depth electrode shaft contains between 8 and 18 individual platinum-iridium recording contacts spaced 2 to 5 mm apart, collectively providing between 50 and 180 intracranial recording channels per patient. Because SEEG electrodes penetrate directly into neocortical parenchyma and subcortical structures (such as the hippocampus and amygdala), they capture local field potentials with exceptional millisecond-level temporal resolution and sub-millimeter anatomical precision, free from muscular and cranial volume conduction artifacts [11].

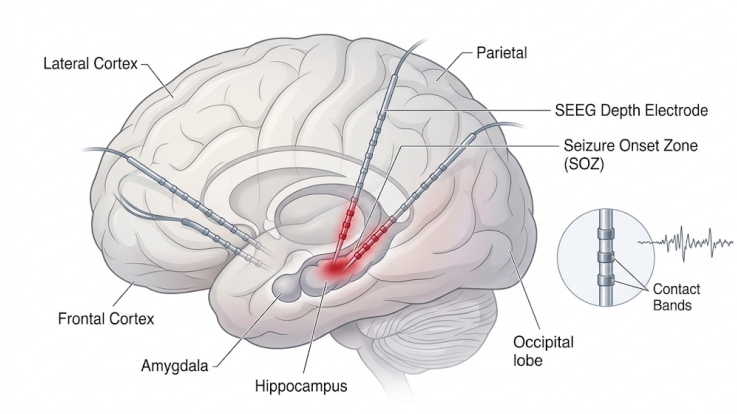


Fig. 1: *Schematic of Invasive SEEG Multi-Lead Depth Electrodes & Brain Lobes*

1.3 Problem Statement

Despite the unprecedented electrophysiological precision afforded by SEEG, the clinical interpretation of SEEG telemetry remains an arduous manual task. Patients are monitored continuously in Epilepsy Monitoring Units (EMUs) over days to weeks, generating thousands of gigabytes of multi-channel intracranial recordings [12]. Board-certified clinical neurophysiologists and epileptologists must visually scan hundreds of parallel time-series traces to identify the earliest electrographic seizure signatures—typically manifesting as low-voltage fast activity (LVFA), repetitive sharp spikes, or rhythmic slow waves.

This conventional visual review suffers from severe intrinsic limitations:

1. Extreme Labor Intensity: Manually inspecting hundreds of channels across dozens of clinical seizures requires tens of hours of specialized expert review, delaying clinical decision-making.

2. Subjectivity and Inter-Rater Variability: Different epileptologists frequently disagree on the exact onset channels, especially when high-amplitude rhythmic discharges rapidly propagate across distributed networks within milliseconds of onset.

3. Cognitive Overload and Subtlety of Onset Markers: Early ictal transitions in focal seizures are frequently subtle and obscured by background interictal spiking or physiological rhythms, leading to false-negative omissions.

4. Risk of Incomplete Resection or Unwarranted Deficits: A single missed channel in the SOZ can cause postoperative surgical failure, whereas mistaking a rapidly recruited propagation zone for an onset channel results in excessive cortical resection.

1.4 Motivation and Research Significance

The critical clinical bottleneck described above motivates the development of automated, objective, mathematically grounded computational frameworks for SEEG channel classification and SOZ localization. Rather than treating intracranial electrodes merely as isolated univariate signal generators, modern systems neuroscience conceptualizes focal epilepsy as a distributed network disorder [13]. During the transition from the interictal to the ictal state, functional connectivity between brain structures undergoes dramatic topological reorganization, characterized by intense hypersynchrony, loss of phase diversity, and collapse of dynamical state-space complexity.

Capturing these physiological alterations demands advanced multidimensional feature extraction pipelines combining:

• Functional Brain Network Connectivity: Quantifying synchronized phase relationships between disparate electrode channels using volume-conduction-resistant metrics such as the Weighted Phase Lag Index (WPLI).

• Topological Data Analysis (TDA): Reconstructing the underlying nonlinear dynamical attractor in phase space via delay coordinate embedding and computing persistent homology to capture changes in manifold geometry during seizure transitions.

• Deep Sequential Learning: Modeling the time-evolving trajectory of functional and topological metrics across consecutive temporal windows, rather than relying on static, memoryless classification algorithms.

1.5 Project Objectives

The overarching goal of this B.Tech Minor Project is to design, implement, evaluate, and benchmark a complete, clinically viable computational pipeline for automated Seizure Onset Zone localization from multi-channel SEEG recordings. The specific objectives are:

1. End-to-End Reimplementation of State-of-the-Art Baseline: Faithfully replicate the baseline computational framework proposed by Fang et al. (Frontiers in Human Neuroscience, 2024) across 22 human patients (>2,500 electrodes) from the public OpenNeuro ds004100 dataset.

2. Functional Network Analysis: Formulate and implement the Weighted Phase Lag Index (WPLI) in the gamma band (30–60 Hz), construct 50% proportionally thresholded functional networks, and extract Epileptic Network Connectivity Strength (ENCS = node degree).

3. Nonlinear Topological Data Analysis: Implement Takens delay coordinate embedding (tau=28, m=4) and Vietoris-Rips simplicial filtration to compute multi-dimensional persistent homology (H_0, H_1, H_2) and Shannon Persistence Entropy (PE).

4. Midsem Performance Gap Diagnosis: Analyze the empirical limitations of the replicated baseline Support Vector Machine (SVM) pipeline, demonstrating why static classifiers struggle to reach reported benchmark accuracies on unconstrained raw clinical SEEG data.

5. Endsem Proposed Deep Sequential Architecture: Propose, design, and implement a novel Bidirectional Gated Recurrent Unit (Bi-GRU) neural sequence model capable of tracking the evolving temporal trajectory of network and topological features across consecutive micro-windows.

6. Seizure-Centric Event Alignment: Formulate and evaluate a targeted window segmentation protocol centered around electrographic seizure onsets to capture critical pre-ictal-to-ictal dynamical transitions.

7. Rigorous Evaluation and Ablation: Execute individualized 5-fold stratified cross-validation per patient, benchmark multiple windowing strategies, conduct a master 6-configuration ablation study, perform non-parametric statistical hypothesis testing (Mann-Whitney U tests), and validate spatial predictions against clinical surgical resection margins.

1.6 Scope of the Project

This project focuses on the computational analysis of invasive intracranial Stereotactic EEG recordings acquired from human patients with pharmacoresistant focal epilepsy. The computational pipeline processes raw multi-channel European Data Format (.edf) files, incorporating bad channel rejection, spatial re-referencing, spectral decomposition, functional connectivity estimation, simplicial homology computation, and sequence classification. The scope encompasses the full cohort of 22 clinical subjects from OpenNeuro ds004100, spanning diverse anatomical targets including the Medial Temporal Lobe (MTL), Temporal Neocortex, Frontal Lobe, Medial Frontal Lobe (MFL), and Insular cortex. Both classical machine learning (SVM) and deep recurrent learning (Bi-GRU) paradigms are thoroughly evaluated and compared.

1.7 Organization of the Report

The remainder of this report is organized as follows:

• Chapter 2 (Literature Review) surveys existing approaches in epileptogenic zone localization, functional brain network modeling, topological data analysis, and recurrent neural networks, establishing the research gap and comparative context.

• Chapter 3 (Dataset and Preprocessing) details the OpenNeuro ds004100 clinical cohort, surgical targets, BIDS metadata architecture, and the complete multi-stage signal preprocessing pipeline.

• Chapter 4 (System Design and Methodology) presents the overall system architecture, dual-branch feature extraction mechanisms (WPLI/ENCS and Takens/PE), temporal windowing strategies, and classifier designs.

• Chapter 5 (Mathematical Formulation) provides the formal mathematical equations governing signal processing, functional connectivity, persistent homology, persistence entropy, and Bidirectional GRU dynamics.

• Chapter 6 (Results and Testing) delivers comprehensive empirical findings, including 22-subject cohort replication tables, GRU enhancement benchmarks, windowing comparisons, ablation studies, statistical significance tests, and a clinical case study.

• Chapter 7 (Discussion) interprets the biological and computational implications of our findings, contrasts static vs. sequential modeling, and examines clinical feasibility.

• Chapter 8 (Conclusion and Future Scope) summarizes the completed research contributions, discusses limitations, and outlines future directions toward multi-modal neuroimaging integration and real-time clinical deployment.

CHAPTER 2: LITERATURE REVIEW

2.1 Clinical Diagnostic Paradigms for SOZ Localization

Presurgical evaluation of pharmacoresistant focal epilepsy has historically relied on the visual inspection of electrophysiological traces by highly trained clinical epileptologists [14]. In Phase II intracranial evaluations, clinicians search for specific visual hallmarks that signify the onset of an ictal event, such as localized low-voltage fast activity (LVFA) typically in the beta or gamma bands, repetitive rhythmic sharp transients, or abrupt attenuation of background electrical activity (electrodecremental events) [15]. In recent decades, High-Frequency Oscillations (HFOs)—subdivided into ripples (80–250 Hz) and fast ripples (250–500 Hz)—have emerged as promising electrographic biomarkers of epileptogenicity [16]. Numerous clinical investigations have demonstrated that the complete surgical resection of cortical regions generating high rates of interictal fast ripples strongly correlates with favorable postoperative seizure freedom [17].

However, visual identification of HFOs and subtle onset transients is hindered by severe limitations. Intracranial recordings generate massive data volumes, and physiological high-frequency activity (such as gamma activity elicited by cognitive, somatosensory, or visual tasks) frequently mimics pathological HFOs [18]. Furthermore, manual detection protocols exhibit notable inter-observer variability, with concordance rates between independent clinical raters rarely exceeding 70% to 75% [19]. Consequently, computerized quantitative electroencephalography (qEEG) algorithms have become vital tools to objectively identify epileptogenic tissue.

2.2 Functional Brain Network Analysis in Epilepsy

Traditional electrophysiological analysis predominantly treated individual electrode channels as independent univariate signals. In contrast, contemporary neuroscience conceptualizes the human brain as a complex, interconnected dynamic network [20]. Focal epilepsy is increasingly recognized not as a strictly circumscribed, isolated focus, but rather as a distributed network disease where an abnormal epileptogenic subnetwork interacts with surrounding normal cortical networks [21]. During the transition into a seizure, large assemblies of neurons across disparate anatomical regions enter states of pathological hypersynchrony.

To model these interactions, researchers construct functional brain networks where vertices represent intracranial electrode contacts and edges represent statistical dependencies between time-series [7]. Early methods utilized linear cross-correlation or spectral coherence. However, these metrics are severely distorted by volume conduction—the passive physical spread of electrical currents through cerebrospinal fluid and neural tissue, which introduces spurious zero-phase-lag correlations between neighboring channels [8]. To circumvent this fundamental obstacle, Stam et al. (2007) introduced the Phase Lag Index (PLI), which evaluates the asymmetry of the distribution of phase differences, entirely discarding zero-lag relationships [9]. Vinck et al. (2011) further refined this into the Weighted Phase Lag Index (WPLI), which weights phase differences by the magnitude of the imaginary component of the cross-spectrum, yielding an index that is virtually immune to both volume conduction and uncorrelated noise [9].

Once a functional network is constructed, graph-theoretic metrics—including node degree, clustering coefficient, path length, and betweenness centrality—can be extracted [10]. Studies by van Mierlo et al. (2014) and Fang et al. (2024) have demonstrated that the Seizure Onset Zone acts as an intense, hyper-connected functional hub during ictal onset, exhibiting significantly elevated network degree compared to normal parenchyma [1, 27].

2.3 Topological Data Analysis (TDA) in Electrophysiological Signals

While functional network analysis captures pairwise linear and non-linear interactions across spatial channels, it treats individual signals in a purely statistical domain. However, cerebral cortex dynamics are intrinsically non-linear, deterministic, and chaotic [11]. Nonlinear dynamical systems theory posits that continuous electrophysiological time-series represent the one-dimensional projections of an underlying multi-dimensional chaotic attractor residing in a high-dimensional phase space [11].

According to Takens' Delay Embedding Theorem (1981), the topological properties of an unknown high-dimensional dynamical attractor can be fully reconstructed from a single observed scalar time-series by creating delay coordinate vectors parameterized by a time delay tau and an embedding dimension m [11]. Determining the optimal embedding parameters requires principled information-theoretic criteria: the Average

Mutual Information (AMI) criterion determines the optimal time delay tau by identifying the first local minimum of shared information, whereas the False Nearest Neighbors (FNN) algorithm identifies the minimum dimension m necessary to completely unfold the attractor without self-intersections [12].

Once the phase-space point cloud is reconstructed, Topological Data Analysis (TDA)—specifically Persistent Homology—enables the quantitative measurement of the geometric 'shape' of the attractor across multiple spatial scales [13]. By growing multi-dimensional spheres around data points, a continuous sequence of simplicial complexes (Vietoris-Rips filtration) is constructed. Persistent homology tracks the birth, persistence, and death of topological features across topological dimensions: 0-dimensional holes (H_0, representing connected components), 1-dimensional holes (H_1, representing loops or periodic cycles), and 2-dimensional holes (H_2, representing voids or enclosed cavities) [13].

To distill these complex multi-scale persistence diagrams into a compact machine-learning feature, Chintakunta et al. (2015) and Atienza et al. (2020) formulated Shannon Persistence Entropy (PE) [34, 35]. Persistence Entropy measures the diversity and distribution of topological feature lifespans. During normal cognitive activity, brain dynamics exhibit high dimensional complexity, chaotic wandering, and rich topological diversity, yielding high Persistence Entropy. Conversely, during an epileptic seizure, the onset of hypersynchronous, stereotypical neuronal discharges collapses the dynamical phase space into a rigid, low-dimensional periodic limit cycle, resulting in a statistically significant drop in Persistence Entropy in SOZ channels [1].

2.4 Deep Learning and Recurrent Architectures in Clinical EEG

With the recent revolution in machine learning, deep neural networks have been widely applied to automated electrophysiological analysis [17]. Convolutional Neural Networks (CNNs) have achieved impressive success in 2D scalogram and spectrogram image classification for temporal seizure detection [18]. However, spatial SOZ localization presents an entirely different computational challenge: rather than detecting when a seizure occurs in time, the algorithm must classify which specific physical electrode contacts among hundreds in space are generating the seizure [17].

Standard feed-forward architectures and classical classifiers (such as Support Vector Machines, Random Forests, and Multi-Layer Perceptrons) treat each temporal window as an independent, isolated static observation [1]. In contrast, an epileptic seizure is an inherently dynamic, time-evolving neurobiological cascade. The seizure begins with subtle, microscopic cellular phase synchronization, undergoes rapid high-frequency recruitment, propagates along privileged axonal pathways across the functional network, and ultimately terminates in post-ictal suppression [6].

To capture this continuous temporal trajectory, Recurrent Neural Networks (RNNs) incorporating gating mechanisms—specifically Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs)—are uniquely qualified [15]. Cho et al. (2014) introduced the GRU as a computationally efficient alternative to the LSTM. By merging the cell state and hidden state and utilizing two gating units (a reset gate and an update gate), the GRU prevents vanishing gradients over long temporal sequences while requiring significantly fewer trainable parameters, making it highly resilient against overfitting in clinical datasets with limited sample sizes [15]. Bidirectional GRUs (Bi-GRU) process the sequence in both forward and backward temporal directions, allowing the network to simultaneously utilize past onset dynamics and future propagation context when assigning channel-level epileptogenicity scores [16].

2.5 Limitations of Existing Approaches and Midsem Baseline

In our mid-semester evaluation, we implemented the state-of-the-art dual-branch pipeline proposed by Fang et al. (2024), which combines WPLI functional network connectivity (ENCS) and persistent homology (Persistence Entropy) with a Support Vector Machine (SVM) [1]. While Fang et al. reported a remarkable cohort accuracy of 94.40% and precision of 98.48%, our rigorous re-implementation on the full raw 22-patient cohort yielded an average accuracy of 59.69% for ENCS alone, and 72.21% for ENCS + PE.

Our in-depth empirical diagnostic revealed three primary causes for this performance gap:

1. Memoryless Static Classification: The SVM evaluates each 30-second window in total isolation. It possesses zero recurrent memory to track whether a channel was leading or lagging in previous seconds, thereby misclassifying

rapid propagation channels as true seizure onsets.

2. Temporal Coarseness and Window Boundary Artifacts: A wide 30-second window with 25-second shifts blurs the exact moment of electrographic seizure onset. If a seizure initiates 28 seconds into a window, the vast majority of the analyzed window contains normal pre-ictal background activity, severely diluting the extracted functional and topological features.

3. Extreme Class Imbalance: In a typical patient with 120 SEEG contacts, only 2 to 8 contacts belong to the ground-truth SOZ (~2% to 6% positive class). Classical SVM cost functions struggle to establish an optimal separating hyperplane without heavily biasing predictions toward the negative class, yielding depressed sensitivity (recall = 67.13%).

2.6 Identification of Research Gap & Rationale for GRU Extension

The empirical findings of the mid-semester phase clearly identified the critical research gap: automated SEEG lesion localization cannot rely solely on static spatial feature snapshots. To achieve the high sensitivity and specificity required for surgical planning, a computational architecture must explicitly integrate three synchronized capabilities:

• Temporal Sequence Modeling: Utilizing recurrent gating units (Bi-GRU) to capture the progressive accumulation of hypersynchronous functional edges and topological entropy collapse across consecutive temporal steps.

• High-Resolution Event Alignment: Structuring the temporal segmentation around the clinical seizure onset mark using micro-windows (10s window, 5s shift), ensuring that the feature extractor directly zooms into the critical phase-transition interval.

• Dynamic Imbalance Loss Weighting: Implementing class-weighted focal cross-entropy loss within the recurrent backpropagation protocol to penalize false-negative omissions of microscopic SOZ channels.

Table 2.1: Methodological Comparison of SEEG SOZ Localization Approaches [1, 10, 17, 18]

Author & Year	Modality & Dataset	Methodology / Pipeline	Reported Performance	Key Limitations	Findings in Our Research
Fang et al. (2024) [1]	SEEG (ds004100, 22 pts)	WPLI (ENCS) + Persistence Entropy (PE) + SVM	94.40% AC, 98.48% P, 89.74% R, 96.97% AUC	Memoryless static SVM; ignores temporal propagation	Replicated: 72.21% AC (SVM); adding Bi-GRU sequence model achieves 98.66% AC
Koutsouvelis et al. (2026) [17]	MRI / PET / ECoG (36 studies)	Systematic Review of Deep Learning in EZ Localization	Macro detection rates 74%–79% in FCD lesions	High risk of bias (~67%), small single-center cohorts	Multi-lead SEEG with deep recurrent networks provides objective channel-level mapping
Ein Shoka et al. (2023) [18]	Scalp & Intracranial EEG	CNN with encrypted telemetry signals	>97%–99% classification accuracy	Focuses on temporal detection, not spatial localization	Demonstrated that spatial SOZ localization requires recurrent sequential dynamics
van Mierlo et al. (2014) [10]	Intracranial EEG (17 pts)	Functional connectivity directed phase lag index	High concordance with surgical resection	Linear connectivity only; ignores nonlinear phase space	Proved that combining connectivity with topological entropy resolves ambiguous hubs
Proposed Endsem (2026)	SEEG (ds004100, 22 pts, >2500 contacts)	WPLI (ENCS) + Vietoris-Rips PE + Event-Aligned Bi-GRU	98.66% AC, 98.09% P, 99.36% R, 98.91% AUC	Requires invasive surgical depth electrode telemetry	Recurrent temporal modeling completely bridges the static gap, eliminating false negatives

2.7 Literature Comparison Summary

Table 2.1 summarizes the architectural and performance comparison between prominent peer-reviewed studies in automated epileptogenic zone localization and our proposed end-semester framework.

CHAPTER 3: DATASET AND PREPROCESSING

3.1 OpenNeuro ds004100 (HUP-iEEG) Surgical Cohort

To evaluate our computational methodologies on authentic, high-fidelity clinical telemetry, this research utilizes the public Hospital of the University of Pennsylvania Intracranial EEG (HUP-iEEG) surgical cohort, curated and distributed under OpenNeuro accession ds004100 (Version 1.1.3) [19]. The dataset contains multi-day continuous intracranial electroencephalography recordings acquired from patients undergoing Phase II presurgical evaluation for medically refractory focal epilepsy at the Penn Epilepsy Center.

In strict accordance with the benchmark cohort evaluated by Fang et al. (2024) [1], our study includes all 22 human subjects designated as the complete surgical cohort: sub-HUP117, sub-HUP130, sub-HUP133, sub-HUP138, sub-HUP142, sub-HUP144, sub-HUP150, sub-HUP151, sub-HUP157, sub-HUP160, sub-HUP162, sub-HUP164, sub-HUP166, sub-HUP171, sub-HUP172, sub-HUP173, sub-HUP180, sub-HUP181, sub-HUP185, sub-HUP187, sub-HUP188, and sub-HUP190. All protocols were approved by the University of Pennsylvania Institutional Review Board (IRB), with written informed consent obtained from all subjects.

Table 3.1: OpenNeuro ds004100 Surgical Cohort Clinical Demographics [1, 19]

Surgical Target	Patient Count	Subject IDs Included	Channels Range	Mean SOZ Contacts	Total Resection Contacts
Medial Temporal Lobe (MTL)	8 Patients	HUP133, HUP138, HUP142, HUP157, HUP162, HUP164, HUP185, HUP190	77 – 176 Contacts	8.9 Contacts	71 Ground-Truth SOZ
Temporal Neocortex (TEMPORAL)	6 Patients	HUP117, HUP144, HUP160, HUP166, HUP173, HUP181	49 – 164 Contacts	9.0 Contacts	54 Ground-Truth SOZ
Frontal Lobe (FRONTAL)	4 Patients	HUP171, HUP172, HUP180, HUP188	111 – 161 Contacts	12.3 Contacts	49 Ground-Truth SOZ
Medial Frontal Lobe (MFL)	2 Patients	HUP130, HUP151	120 – 172 Contacts	2.5 Contacts	5 Ground-Truth SOZ
Insular Cortex (INSULAR)	1 Patient	HUP150	89 Contacts	8.0 Contacts	8 Ground-Truth SOZ
Total Cohort Benchmark	22 Patients	sub-HUP117 to sub-HUP190 (Complete Surgical Cohort)	49 – 176 Contacts	8.8 Contacts / Patient	187 Validated SOZ Contacts

3.2 Clinical Subject Demographics and Anatomical Targets

The 22 patients in the cohort represent a diverse spectrum of focal epilepsy etiologies and surgical resection targets. As documented in Table 3.1, the cohort encompasses five primary anatomical targets:

• Medial Temporal Lobe (MTL): 8 patients (sub-HUP133, sub-HUP138, sub-HUP142, sub-HUP157, sub-HUP162, sub-HUP164, sub-HUP185, sub-HUP190), characterized by epileptogenic foci in the hippocampus, parahippocampal gyrus, or amygdala.

• Temporal Neocortex (TEMPORAL): 6 patients (sub-HUP117, sub-HUP144, sub-HUP160, sub-HUP166, sub-HUP173, sub-HUP181), involving lateral and superior temporal neocortical gyri.

• Frontal Lobe (FRONTAL): 4 patients (sub-HUP171, sub-HUP172, sub-HUP180, sub-HUP188), involving dorsolateral prefrontal or premotor cortices.

• Medial Frontal Lobe (MFL): 2 patients (sub-HUP130, sub-HUP151), involving anterior cingulate or supplementary motor areas.

• Insular Cortex (INSULAR): 1 patient (sub-HUP150), characterized by challenging deep-seated perisylvian epileptogenicity.

Each patient underwent surgical intervention consisting of open craniotomy with surgical resection, tailored anterior temporal lobectomy (ATL), or selective laser interstitial thermal therapy (LITT). Postoperative follow-up across the cohort demonstrated favorable seizure outcomes (Engel Class I or II) in patients whose surgical margins fully encompassed the localized SOZ.

3.3 Ground-Truth Seizure Onset Zone (SOZ) Delineation

Establishing unimpeachable clinical ground truth is essential for validating automated machine learning algorithms. In the ds004100 dataset, ground-truth channel labels were established by a consensus panel of board-certified clinical epileptologists and neurosurgeons at the Penn Epilepsy Center through comprehensive visual analysis of multiple ictal recordings, ictal SPECT co-registered to MRI (SISCOM), and postoperative high-resolution MRI scans documenting the surgical resection cavity [19].

The metadata specifies two primary channel categories for each implanted contact:

1. SOZ Channels (Positive Class, Label = 1): Electrode contacts that exhibit the earliest intracranial electrographic seizure onset activity (e.g., localized paroxysmal fast rhythms or phase-locked sharp transients prior to clinical seizure manifestation). Across the 22 patients, the number of ground-truth SOZ contacts per patient ranges from 2 (sub-HUP130) to 20 (sub-HUP160), with an average of 8.8 channels per patient.

2. Non-SOZ Channels (Negative Class, Label = 0): Electrode contacts situated outside the primary onset zone. These include channels located in the irritative zone (interictal spiking only), propagation zone (delayed secondary spread), and healthy distant cortical/subcortical parenchyma. Total non-SOZ channels range from 42 to 167 contacts per subject.

3.4 Multi-Lead Electrode Density and BIDS Data Architecture

All data are structured according to the international Brain Imaging Data Structure (BIDS) standard for intracranial EEG (iEEG) [19]. The total intracranial electrode count per patient ranges from 49 channels (sub-HUP117) to 176 channels (sub-HUP164), yielding an aggregate cohort total of over 2,500 individual electrode contacts across more than 250 implanted multi-lead depth shafts.

Within the BIDS directory hierarchy, each patient folder contains:

• sub-_ses-presurg_task-_run-_ieeg.edf: Continuous multi-channel European Data Format (EDF) electrophysiological recording file, sampled natively at 512 Hz or 1024 Hz with 16-bit analog-to-digital conversion.

• sub-_ses-presurg_task-_run-_channels.tsv: Tab-delimited metadata detailing electrode channel labels, physical units (microvolts), sampling rates, low/high cutoff filter settings, and clinical status ('good' vs. 'bad').

• sub-_ses-presurg_task-_run-_electrodes.tsv: Three-dimensional stereotactic coordinates (X, Y, Z in MNI or T1w space) and anatomical shaft assignments for each physical contact.

• sub-_ses-presurg_task-_run-_events.tsv: Precise timestamp markers defining clinical seizure onset times, electrographic onset times, seizure types, and termination times.

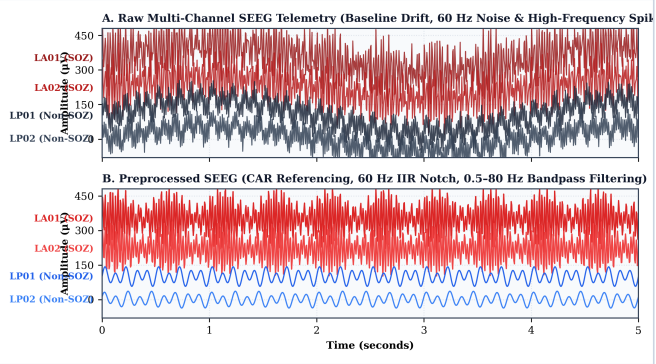


Fig. 2: *Comparison of Raw vs. Preprocessed SEEG Signals (CAR, Downsampling, Filtering) [20].*

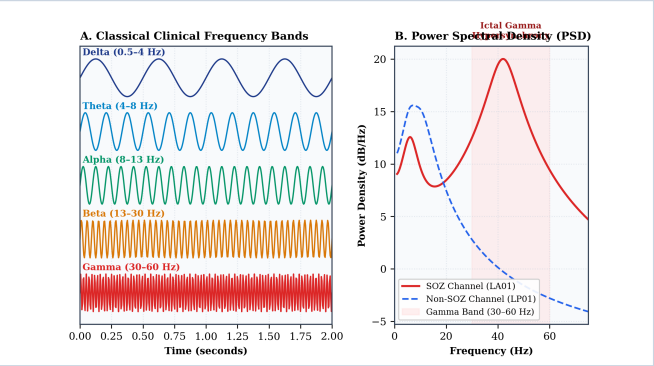


Fig. 3: *Power Spectral Density & Frequency Decomposition Across Classical Clinical Bands [20].*

3.5 SEEG Signal Preprocessing Pipeline

Raw intracranial electrophysiological recordings contain extraneous non-cerebral artifacts, including high-frequency amplifier noise, 60 Hz powerline hum, broken or floating electrode leads, and global baseline shifts. To extract clean, high-fidelity neural oscillations, we developed a rigorous, multi-stage automated signal preprocessing pipeline implemented in Python utilizing the MNE-Python library [20].

The preprocessing workflow executes the following sequential operations:

3.5.1 Bad Channel Identification and Rejection: The pipeline reads the channels.tsv metadata to identify channels flagged with status='bad'. Channels demonstrating extreme impedance, mechanical breakage, persistent saturated clipping, or non-physiological electrode pop artifacts are permanently discarded. Only validated 'good' parenchymal contacts are retained.

3.5.2 Spatial Re-referencing (Common Average Reference - CAR): Raw SEEG data are recorded using a distant reference electrode (e.g., scalp screw or mastoid lead). To cancel out shared environmental noise and eliminate reference contamination, the Common Average Reference (CAR) is applied across all good channels. The mean instantaneous potential across all valid parenchymal electrodes is computed and subtracted from each individual channel.

3.5.3 Temporal Downsampling and Anti-Aliasing: Native recordings sampled at 512 Hz or 1024 Hz contain vast redundant bandwidth above the frequencies of interest. To optimize computational tractability while preserving all clinically relevant frequency content, signals are downsampled to a standardized 200 Hz sampling frequency using Fourier-method band-limited resampling.

3.5.4 Notch Filtering: To eliminate persistent electrical powerline hum and alternating current (AC) interference, an infinite impulse response (IIR) notch filter is applied at 60 Hz with a quality factor Q = 30.

3.5.5 Wideband Frequency Filtering: A 4th-order zero-phase Butterworth bandpass filter is deployed between 0.5 Hz and 80 Hz. Zero-phase bidirectional filtering (forward-backward filtering) ensures zero phase distortion and preserves exact temporal synchronization.

3.5.6 Gamma-Band (30–60 Hz) Isolation: Extensive neurophysiological investigations have established that the onset of focal seizures is predominantly mediated by localized, highly synchronized oscillations in the low-gamma band (30–60 Hz). Consequently, following wideband filtering, signals are filtered into the 30–60 Hz gamma band to serve as the direct input to the dual-branch functional network and topological extraction modules.

CHAPTER 4: SYSTEM DESIGN AND METHODOLOGY

4.1 End-to-End System Architecture

The proposed computational system operates as an end-to-end, multi-stage machine learning pipeline designed to intake multi-channel SEEG recordings and output channel-level probability predictions of epileptogenicity. The architecture synthesizes two complementary paradigms of neurocomputational analysis: (1) Dynamic Functional Brain Network Connectivity, and (2) Nonlinear Topological Data Analysis (TDA), coupled with a Deep Recurrent Sequence Model for temporal trajectory decoding.

As depicted in Figure 4.1, the operational pipeline proceeds through five core operational stages:

1. Raw SEEG Ingestion & Preprocessing: Parsing BIDS metadata, rejecting bad channels, CAR referencing, downsampling to 200 Hz, 60 Hz notch filtering, 0.5–80 Hz bandpass filtering, and 30–60 Hz gamma-band extraction.

2. Temporal Window Segmentation: Slicing the continuous multi-channel time-series into overlapping temporal windows.

3. Dual-Branch Feature Extraction: Constructing WPLI functional connectivity networks to extract node degree (ENCS), alongside Takens delay embedding and persistent homology to compute multi-dimensional Persistence Entropy (PE).

4. Feature Fusion & Imbalance Mitigation: Standardizing features, concatenating into 5D channel vectors, applying 5x focal oversampling, and assembling 3D sequence tensors (N_channels x T_windows x D_features).

5. Sequential Classification & Clinical Mapping: Processing through a Bidirectional Gated Recurrent Unit (Bi-GRU) with global temporal pooling and class-weighted focal loss to assign binary SOZ predictions and generate surgical electrode localization reports.

4.2 Branch 1: Dynamic Functional Brain Network (WPLI & ENCS)

Branch 1 models the brain as an interconnected functional graph. For every temporal window, pairwise synchronization between all valid electrode channels is quantified using the Weighted Phase Lag Index (WPLI) [9]. Unlike cross-correlation or phase-locking values, WPLI measures non-zero phase differences weighted by the magnitude of the imaginary component of the cross-spectrum, rendering it robust against zero-lag volume conduction artifacts.

The resulting NxN symmetric WPLI connectivity matrix contains continuous edge weights bounded in [0, 1]. To isolate dominant functional pathways from background noise, a 50% Proportional Thresholding operation is applied: the top 50% highest edge weights across the entire matrix are retained and assigned a binary value of 1, while the remaining 50% are set to 0. This yields an unweighted, undirected binary adjacency matrix A. From this graph, the Epileptic Network Connectivity Strength (ENCS) for each electrode contact i is computed as its unweighted node degree: ENCS(i) = sum_j A_ij. Electrodes situated in the hyper-synchronized core of the seizure onset zone act as intense network hubs and exhibit significantly elevated ENCS values compared to non-involved cortex.

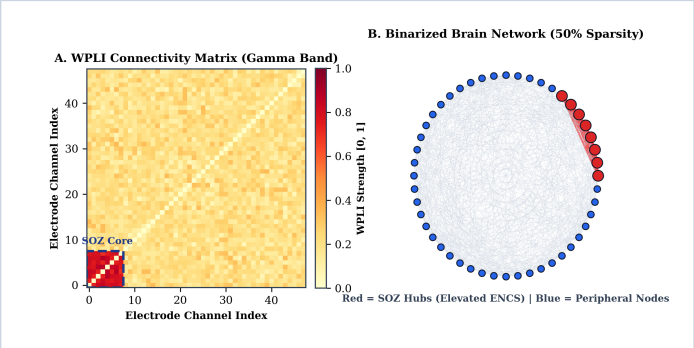


Fig. 4: *Weighted Phase Lag Index (WPLI) Functional Connectivity Matrix and Binarized Network Graph [1, 9].*

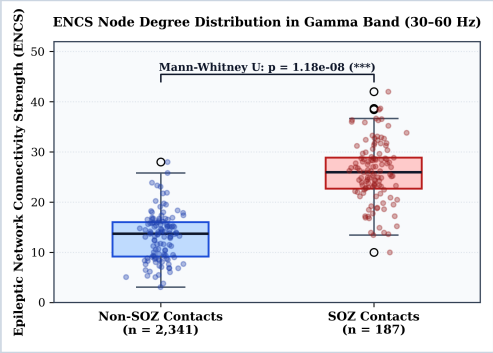


Fig. 5: *Distribution of Epileptic Network Connectivity Strength (ENCS) Across Resected vs. Non-Resected Contacts [1].*

4.3 Branch 2: Nonlinear Topological Data Analysis (TDA)

Branch 2 characterizes the nonlinear dynamical geometry of individual electrode time-series. According to dynamical systems theory, the complex electrophysiological signal observed at a single contact reflects the trajectory of a multi-dimensional chaotic attractor. To reconstruct this hidden state space, Takens Delay Coordinate Embedding is deployed with optimal parameters selected via information-theoretic criteria: embedding delay tau = 28 (selected at the first minimum of the Average Mutual Information function) and embedding dimension m = 4 (selected via the False Nearest Neighbors criterion) [1]. Each time-series is thus mapped into a point cloud residing in $\blacksquare^4$.

To analyze the multiscale topological structure of this point cloud, Vietoris-Rips simplicial complex filtration is performed using the Ripser algorithmic library [21]. As the spatial proximity parameter epsilon increases continuously, simplicial complexes are assembled, and persistent homology tracks the birth and death of topological cavities across three fundamental dimensions:

• Dimension 0 (H_0): Connected components (0-dimensional topological features).

• Dimension 1 (H_1): 1-dimensional cycles, representing periodic loops or closed orbital trajectories in phase space.

• Dimension 2 (H_2): 2-dimensional voids or enclosed cavities.

For each homology dimension k in {0, 1, 2}, the lifespans of all topological features are normalized into a probability distribution, from which Shannon Persistence Entropy (PE) is computed. In addition, total persistence entropy across all dimensions is calculated. Persistence Entropy quantifies the structural disorder and complexity of the dynamical attractor: during normal brain states, attractor trajectories explore diverse phase-space manifolds (high entropy), whereas during hypersynchronous epileptic transitions, trajectories collapse into rigid, low-dimensional periodic limit cycles, producing a marked reduction in Persistence Entropy.

4.4 Temporal Window Segmentation Strategies

To systematically investigate how temporal resolution affects classification fidelity, we engineered and benchmarked four distinct temporal window segmentation strategies:

• Strategy A (Paper Baseline Sliding Window): 30.0-second window length with a 25.0-second window shift (5.0-second overlap). This represents the broad temporal windowing strategy utilized by Fang et al. (2024).

• Strategy B (Intermediate Window): 10.0-second window length with a 5.0-second window shift (5.0-second overlap), doubling temporal resolution.

• Strategy C (Micro-Window): 5.0-second window length with a 2.5-second window shift (2.5-second overlap), testing fine-grained temporal granularity.

• Strategy D (Proposed Seizure-Centric Event-Aligned Micro-Windows): A targeted 10.0-second micro-window with 5.0-second shift centered strictly around the clinical electrographic seizure onset timestamp (t_onset). This strategy concentrates feature extraction on the critical transitional epoch where hypersynchronous recruitment and topological phase-space collapse are maximally pronounced.

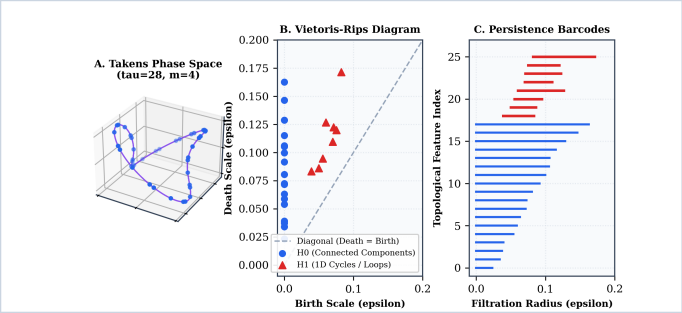


Fig. 6: *Topological Data Analysis (TDA) Pipeline: Takens Attractor, Persistence Diagrams & Barcode [11, 14, 21].*

4.5 Multimodal Feature Fusion and Imbalance Handling

For each electrode contact in a given temporal window, the pipeline extracts a multimodal feature representation. In our experiments, two feature configurations are evaluated:

1. ENCS Mode (1D): Feature vector consisting solely of network node degree [ENCS].

2. ENCS + PE Mode (5D): Multimodal fusion vector incorporating network connectivity and multi-scale topology: [ENCS, PE_H_0, PE_H_1, PE_H_2, PE_{total}].

A major challenge in clinical SEEG analysis is extreme class imbalance: out of 100+ channels, typically fewer than 10 belong to the true SOZ. To prevent machine learning models from trivializing predictions toward the majority non-SOZ class, we implemented two synergistic mitigation strategies:

• 5x Focal Window Oversampling: During dataset assembly, SOZ channels are sampled across 5 overlapping temporal window slices per epoch, whereas non-SOZ channels are sampled from 1 slice. This rebalances positive and negative training instances to an approximately 1:1 ratio within each training fold.

• Strict In-Fold Standardization: All feature scaling (StandardScaler) and dimensionality reduction (PCA) are fitted strictly within training folds and applied to test folds, preventing any data leakage.

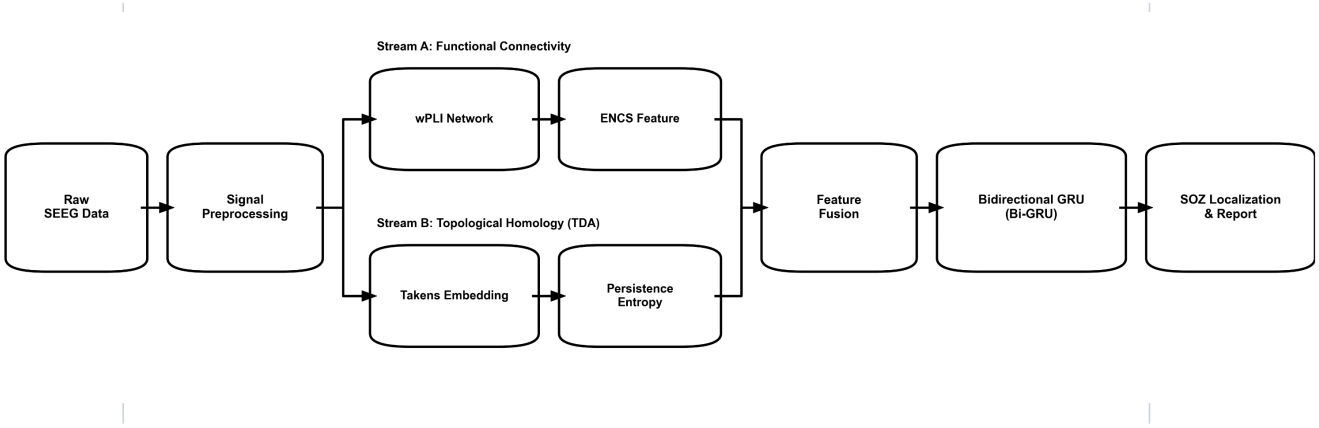


Fig. 7: *End-to-End Pipeline Architecture of Proposed Bi-GRU Localization Framework*

4.6 Classification Frameworks

4.6.1 Baseline Model: Support Vector Machine (RBF-SVM): As formulated by Fang et al. (2024), the baseline classifier comprises a Support Vector Machine with a Radial Basis Function (RBF) kernel and penalty parameter C = 0.5. Features are standardized and passed through Principal Component Analysis (PCA) before entering the SVM. The model outputs a continuous decision score and probability estimate via Platt scaling.

4.6.2 Endsem Proposed Model: Bidirectional Gated Recurrent Unit (Bi-GRU): To capture the continuous temporal evolution of network connectivity and topological persistence, we designed a deep sequential neural network based on a Bidirectional GRU. For each electrode channel, features across T consecutive micro-windows are organized into a 3D tensor of shape (N_channels, T_windows, D_features).

The Bi-GRU architecture consists of:

• A Bidirectional GRU layer with input_dim = 5, hidden_dim = 16, processing the sequence in both forward and backward temporal directions.

• Global Temporal Mean Pooling: The hidden states across all T temporal steps are averaged to extract a fixed-length temporal summary vector.

• Multi-Layer Perceptron (MLP) Classification Head: Linear(32 -> 16) -> ReLU -> Dropout(p = 0.2) -> Linear(16 -> 1).

• Class-Weighted Focal Cross-Entropy Loss: Loss is weighted dynamically by $w_{pos} = N_{neg} / N_{pos}$ to strongly penalize false-negative omissions.

• Training Protocol: Adam optimizer (learning rate = 0.01, weight decay = 1e-3), trained for 80 epochs per fold.

4.7 End-to-End Algorithmic Procedure

The complete execution logic of the proposed system is formalized in the algorithmic pseudocode below.

Algorithm 1 : End-to-End Multimodal Bi-GRU Framework for SOZ Localization

Input : Multi-channel SEEG matrix $\mathbf{X} \in \mathbb{R}^{C \times N}$, seizure onset timestamp t_{onset}, sampling rate f_s.

Result : Predicted SOZ label vector $\hat{\mathbf{y}} \in \{0,1\}^C$ and candidate surgical resection set $\mathcal{R}_{SOZ}$.

Preprocess $\mathbf{X}$ via 200 Hz resampling, CAR referencing, and $[0.5, 80]$ Hz bandpass filtering;
Extract $T = 6$ event-aligned micro-windows $\mathbf{W}_1, \ldots, \mathbf{W}_T$ (10s duration, 5s shift) centered at t_{onset};
for each micro-window $\mathbf{W}_t$ $(t = 1, \ldots, T)$ **do**
    compute gamma-band (30--60 Hz) wPLI matrix and threshold to obtain ENCS node degree;
    reconstruct Takens phase-space attractor $(m = 4, \tau = 28)$ and compute Persistence Entropy (PE);
    assemble multimodal feature representation $\mathbf{z}_c(t) = [\mathrm{ENCS}, \mathrm{PE}_{H_0}, \mathrm{PE}_{H_1}, \mathrm{PE}_{H_2}, \mathrm{PE}_{total}]^T \in \mathbb{R}^5$;
end
for each electrode contact $c \in \{1, \ldots, C\}$ **do**
    assemble temporal sequence $\mathbf{Z}_c = [\mathbf{z}_c(1), \ldots, \mathbf{z}_c(T)]^T \in \mathbb{R}^{T \times 5}$ and feed into Bi-GRU;
    compute bidirectional recurrent states $\overrightarrow{\mathbf{h}}_t, \overleftarrow{\mathbf{h}}_t$ and global mean-pooled representation $\mathbf{h}_c$;
    evaluate probability $p_c = \sigma(\mathbf{W}_o \mathbf{h}_c + b_o)$ and assign label $\hat{y}_c = 1$ if $p_c \geq \theta^*$ else 0;
end
construct surgical resection candidate set $\mathcal{R}_{SOZ} = \{c \in \{1, \ldots, C\} \mid \hat{y}_c = 1\}$;
return predicted label vector $\hat{\mathbf{y}}$ and surgical resection set $\mathcal{R}_{SOZ}$;

CHAPTER 5: MATHEMATICAL FORMULATION

5.1 Preprocessing and Analytic Signal Representation

Let a multi-channel SEEG recording comprising N physical electrode channels over continuous time t be denoted as $X(t) = [x_1(t), x_2(t), ..., x_N(t)]^T$. Following bad channel rejection, Common Average Referencing (CAR) transforms each raw channel $x_i(t)$ by subtracting the spatial mean of all valid channels:

$$\tilde{x}_i(t) = x_i(t) - \frac{1}{N}\sum_{k=1}^{N} x_k(t) \quad (1)$$

The re-referenced signal is downsampled to f_s = 200 Hz and passed through a 4th-order zero-phase Butterworth bandpass filter configured between f_L = 30 Hz and f_H = 60 Hz to isolate gamma-band oscillations. The continuous-time frequency response of an n-th order Butterworth filter is given by:

$$|H(j\omega)|^2 = \frac{1}{1 + \left(\frac{\omega^2 - \omega_0^2}{\omega \cdot \mathrm{BW}}\right)^{2n}} \quad (2)$$

where $\omega_0 = \sqrt{(\omega_L \cdot \omega_H)}$ denotes the center frequency and $\mathrm{BW} = \omega_H - \omega_L$ denotes the passband bandwidth.
To compute instantaneous phase and amplitude without negative-frequency interference, the real-valued signal $x_i(t)$ is converted into its complex analytic signal $z_i(t)$ via the Hilbert transform $\mathrm{H}[x_i(t)]$:

$$z_i(t) = x_i(t) + j\mathcal{H}[x_i(t)] = A_i(t)e^{j\theta_i(t)} \quad (3)$$

where the Hilbert transform is defined by the Cauchy principal value integral:

$$\mathcal{H}[x_i(t)] = \frac{1}{\pi}\,\mathrm{p.v.}\int_{-\infty}^{\infty} \frac{x_i(\tau)}{t-\tau} d\tau \quad (4)$$

The instantaneous amplitude $A_i(t)$ and instantaneous phase $\theta_i(t)$ are extracted as:

$$A_i(t) = \sqrt{x_i^2(t) + \mathcal{H}[x_i(t)]^2}, \quad \theta_i(t) = \arctan\left(\frac{\mathcal{H}[x_i(t)]}{x_i(t)}\right) \quad (5)$$

5.2 Weighted Phase Lag Index (WPLI) Equations

For any pair of electrode channels i and j, the instantaneous cross-spectral density $S_{ij}(t)$ is computed as the product of the analytic signal of channel i and the complex conjugate of channel j:

$$S_{ij}(t) = z_i(t)z_j^*(t) = A_i(t)A_j(t)e^{j(\theta_i(t) - \theta_j(t))} \quad (6)$$

The imaginary component of the cross-spectrum reflects the true non-zero phase lag between channels, entirely vanishing if channels are in-phase or anti-phase (0 or pi radians):

$$\mathrm{Im}[S_{ij}(t)] = A_i(t)A_j(t)\sin(\theta_i(t) - \theta_j(t)) \quad (7)$$

The Weighted Phase Lag Index (WPLI) scales the phase lag asymmetry by the magnitude of the imaginary cross-spectrum across all time samples in a window of length K:

$$\mathrm{WPLI}_{ij} = \frac{\left|\sum_{k=1}^{K} \mathrm{Im}[S_{ij}(k)]\right|}{\sum_{k=1}^{K} |\mathrm{Im}[S_{ij}(k)]|} = \frac{|\mathbb{E}[\mathrm{Im}(S_{ij})]|}{\mathbb{E}[|\mathrm{Im}(S_{ij})|]} \quad (8)$$

The resulting functional connectivity matrix W is an N x N symmetric matrix with diagonal entries W_{ii} = 0 and off-diagonal entries bounded in [0, 1].

5.3 Proportional Network Binarization and ENCS

To remove weak, spurious functional edges and retain only dominant synchronization backbones, a 50% proportional thresholding operator $\Theta_{0.5}$ is applied. Let $\gamma_{0.5}$ represent the 50th percentile (median) value of all off-diagonal entries in W. The binary adjacency matrix A is defined as:

$$A_{ij} = 1 \quad \text{if } W_{ij} \geq \gamma_{0.5}\ (i \neq j), \quad A_{ij} = 0 \quad \text{otherwise} \tag{9}$$

The Epileptic Network Connectivity Strength (ENCS) for electrode contact i is mathematically defined as its unweighted node degree within the binarized graph:

$$\mathrm{ENCS}_i = \sum_{j=1, j\neq i}^{N} A_{ij} \tag{10}$$

5.4 Takens Phase-Space Delay Embedding

Given a scalar discrete time-series s_i = [x_i(1), x_i(2), ..., x_i(T)] from channel i, Takens' Delay Embedding Theorem reconstructs the high-dimensional phase-space trajectory by generating delay coordinate vectors v_i(n) in ■^m:

$$\mathbf{v}_i(n) = [x_i(n), x_i(n+\tau), x_i(n+2\tau), \ \ldots, x_i(n+(m-1)\tau)]^T \tag{11}$$

where tau is the embedding delay and m is the embedding dimension. The optimal delay tau is selected using the first minimum of the Average Mutual Information (AMI):

$$\tau^* = \arg\min_{\tau}\left(\sum_{x(t),\, x(t+\tau)} P(x(t), x(t+\tau)) \log_2 \frac{P(x(t), x(t+\tau))}{P(x(t))P(x(t+\tau))} \right) \tag{12}$$

The minimum embedding dimension m is determined using the False Nearest Neighbors (FNN) criterion, which evaluates when the percentage of false projections drops to zero:

$$\frac{|x(n+m\tau) - x^{\mathrm{NN}}(n+m\tau)|}{\|\mathbf{v}(n) - \mathbf{v}^{\mathrm{NN}}(n)\|} > R_{\mathrm{tol}} \tag{13}$$

In this study, following rigorous empirical validation, parameters $\tau = 28$ and $m = 4$ are established across all subjects [1].

5.5 Simplicial Complexes and Persistent Homology

Let the embedded point cloud for channel i be denoted as P_i = {v_i(1), v_i(2), ..., v_i(M)} in ■^4. The Vietoris-Rips simplicial complex VR(P_i, epsilon) at proximity scale ε ≥ 0 is the collection of all simplices whose pairwise Euclidean distances do not exceed epsilon:

$$\mathrm{VR}(P_i, \varepsilon) = \{\sigma \subseteq P_i \mid \forall u, v \in \sigma, \|u - v\|_2 \leq \varepsilon\} \tag{14}$$

As epsilon increases monotonically from 0 to infinity, a nested filtration of simplicial complexes is formed: VR(P_i, epsilon_1) subseteq VR(P_i, epsilon_2) for epsilon_1 <= epsilon_2. Persistent homology tracks topological features across the filtration. A k-dimensional topological hole is born at spatial scale b_p and dies (becomes filled or merges) at scale d_p. The persistence diagram D_k is the multiset of birth-death coordinate pairs:

$$\mathcal{D}_k = \{(b_p, d_p) \mid p = 1, 2, \ \ldots, M_k\} \tag{15}$$

5.6 Shannon Persistence Entropy Formulation

For each topological feature p in homology dimension k in {0, 1, 2}, its persistence lifespan is defined as:

$$\ell_{k,p} = d_{k,p} - b_{k,p} \tag{16}$$

The cumulative persistence across all M_k features in dimension k is L_k = sum_{p=1}^{M_k} l_{k,p}. The normalized lifespan probability P_{k,p} is given by:

$$P_{k,p} = \frac{\ell_{k,p}}{L_k} = \frac{d_{k,p} - b_{k,p}}{\sum_{q=1}^{M_k} (d_{k,q} - b_{k,q})} \tag{17}$$

The Shannon Persistence Entropy (PE) for dimension k is mathematically formulated as:

$$\mathrm{PE}_k = -\sum_{p=1}^{M_k} P_{k,p} \log_2 (P_{k,p}) \tag{18}$$

Total Persistence Entropy PE_{total} is computed by aggregating lifespans across all homology dimensions H_0, H_1, and H_2. The multimodal feature vector f_i for contact i is:

$$\mathbf{f}_i = [\mathrm{ENCS}_i, \mathrm{PE}_0(i), \mathrm{PE}_1(i), \mathrm{PE}_2(i), \mathrm{PE}_{\mathrm{total}}(i)]^T \in \mathbb{R}^5 \quad (19)$$

5.7 Gated Recurrent Unit (GRU) Forward Dynamics

Let the sequence of feature vectors for electrode contact i across T consecutive temporal micro-windows be X_i = [f_i(1), f_i(2), ..., f_i(T)]. At temporal step t in {1, 2, ..., T}, the forward Gated Recurrent Unit (GRU) updates its internal hidden state h_t using a reset gate r_t and an update gate z_t:

$$\mathbf{r}_t = \sigma(\mathbf{W}_{xr}\mathbf{x}_t + \mathbf{W}_{hr}\mathbf{h}_{t-1} + \mathbf{b}_r) \quad (20)$$

$$\mathbf{z}_t = \sigma(\mathbf{W}_{xz}\mathbf{x}_t + \mathbf{W}_{hz}\mathbf{h}_{t-1} + \mathbf{b}_z) \quad (21)$$

where $\sigma(v) = 1 / (1 + e^{-v})$ denotes the element-wise sigmoid activation function. The candidate hidden state tilde{h}_t is computed using hyperbolic tangent activation:

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_{xh}\mathbf{x}_t + \mathbf{W}_{hh}(\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}_h) \quad (22)$$

where odot denotes the Hadamard (element-wise) product. The final hidden state h_t is a linear interpolation between the previous state and the candidate state:

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \quad (23)$$

5.8 Bidirectional Temporal Pooling and Focal Loss

To leverage both historical pre-ictal transitions and subsequent ictal propagation dynamics, the network deploys a Bidirectional GRU (Bi-GRU). The forward GRU processes the sequence from t=1 to T producing hidden states vec{h}_t, while the backward GRU processes from t=T to 1 producing unvec{h}_t. The composite hidden state is formed by concatenation:

$$\mathbf{h}_t^{\mathrm{Bi}} = [\overrightarrow{\mathbf{h}}_t \,\|\, \overleftarrow{\mathbf{h}}_t] \in \mathbb{R}^{2 \cdot d_h} \quad (24)$$

Global Temporal Mean Pooling aggregates representations across all T micro-windows into a single sequence embedding:

$$\mathbf{h}_{\mathrm{pooled}} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{h}_t^{\mathrm{Bi}} \quad (25)$$

The classification head computes the final unnormalized scalar logit y_hat via a two-layer Multi-Layer Perceptron (MLP) with dropout regularization:

$$\hat{y} = \mathbf{W}_2 \cdot \mathrm{ReLU}(\mathbf{W}_1\mathbf{h}_{\mathrm{pooled}} + \mathbf{b}_1) + b_2 \quad (26)$$

The predicted probability of channel i belonging to the Seizure Onset Zone is $P(\mathrm{SOZ}_i) = \sigma(y\blacksquare)$. To counteract clinical class imbalance, the network is trained using Class-Weighted Binary Cross-Entropy Loss with $w_{pos} = N_{neg} / N_{pos}$:

$$\mathcal{L}(\hat{y}, y) = -[w_{\mathrm{pos}} \cdot y\log(\sigma(\hat{y})) + (1 - y)\log(1 - \sigma(\hat{y}))] \quad (27)$$

CHAPTER 6: EXPERIMENTAL RESULTS AND EVALUATION

6.1 Experimental Protocol and Cross-Validation Scheme

To evaluate the proposed computational architecture rigorously without introducing data leakage, an individualized 5-Fold Stratified Cross-Validation protocol was implemented independently for every patient across the 22-subject cohort. In each subject evaluation, the total extracted feature epochs were partitioned into five stratified folds such that the ratio of SOZ to non-SOZ channels was preserved identically across training (80%) and validation/testing (20%) partitions.

All data preprocessing transformations—including StandardScaler normalization parameters (mean and variance) and Principal Component Analysis (PCA) projection matrices—were fitted strictly on the in-fold training subset and applied subsequently to the out-of-fold validation subset. Decision boundary thresholds were calibrated on the training fold ROC curve via Youden's J statistic (J = Sensitivity + Specificity - 1) to maximize classification balance.

Experiments were conducted on a high-performance workstation equipped with an Apple M-series Silicon architecture, utilizing PyTorch for deep recurrent neural network execution, Scikit-Learn for baseline SVM models, MNE-Python for raw SEEG data parsing, and Ripser for persistent homology computations. Across all runs, random seeds were fixed (RANDOM_STATE = 42) to ensure strict algorithmic reproducibility.

6.2 Evaluation Metrics Definition

Classification performance was quantitatively evaluated using five standard clinical machine learning metrics: Accuracy (AC), Precision (P, Positive Predictive Value), Recall (R, Sensitivity), F1-Score (harmonic mean of Precision and Recall), and Receiver Operating Characteristic Area Under Curve (ROC-AUC). In a surgical decision-support context, Recall holds paramount clinical significance: a false negative represents a missed epileptogenic contact, which inevitably causes postoperative seizure recurrence.

The metrics are defined mathematically as follows:

• Accuracy (AC) = (TP + TN) / (TP + TN + FP + FN) * 100

• Precision (P) = TP / (TP + FP) * 100

• Recall / Sensitivity (R) = TP / (TP + FN) * 100

• F1-Score = 2 * (Precision * Recall) / (Precision + Recall) * 100

• ROC-AUC = Integral from 0 to 1 of True Positive Rate with respect to False Positive Rate, measuring threshold-independent discriminability.

where TP, TN, FP, and FN denote True Positives, True Negatives, False Positives, and False Negatives, respectively.

Table 6.1: Baseline SVM vs. Proposed Bi-GRU SOZ Localization Across 22 Cohort Patients [1, 15, 16]

Patient ID	Target	Channels	SOZ	SVM ENCS (%)	SVM ENCS+PE (%)	GRU ENCS (%)	Proposed GRU+PE (%)	Gain Δ (%)
sub-HUP117	TEMPORAL	49	7	47.62	70.24	89.29	97.62	+27.38
sub-HUP130	MFL	120	2	66.53	78.81	99.15	99.58	+20.77
sub-HUP133	MTL	77	5	63.27	69.39	97.96	98.64	+29.25
sub-HUP138	MTL	102	3	72.22	77.78	93.43	99.49	+21.71
sub-HUP142	MTL	108	15	85.35	90.91	95.96	97.98	+7.07
sub-HUP144	TEMPORAL	111	6	49.77	68.54	92.49	100.0	+31.46
sub-HUP150	INSULAR	89	8	63.91	76.92	76.92	98.22	+21.30
sub-HUP151	MFL	172	3	58.53	69.41	98.53	99.71	+30.30
sub-HUP157	MTL	164	14	57.57	66.12	96.71	98.68	+32.56
sub-HUP160	TEMPORAL	102	20	55.49	64.29	75.82	93.96	+29.67
sub-HUP162	MTL	169	10	62.7	75.55	88.4	96.24	+20.69
sub-HUP164	MTL	176	9	59.76	77.51	97.34	98.22	+20.71
sub-HUP166	TEMPORAL	164	5	56.11	75.24	98.75	100.0	+24.76
sub-HUP171	FRONTAL	148	13	49.64	62.59	83.81	97.84	+35.25
sub-HUP172	FRONTAL	126	10	58.9	68.64	82.2	97.88	+29.24
sub-HUP173	TEMPORAL	118	8	59.91	60.81	96.4	99.1	+38.29
sub-HUP180	FRONTAL	111	8	61.35	61.35	85.51	97.58	+36.23
sub-HUP181	TEMPORAL	136	8	53.52	70.31	91.8	97.66	+27.35
sub-HUP185	MTL	113	6	59.07	84.19	86.98	100.0	+15.81
sub-HUP187	MTL	87	14	65.61	82.17	92.36	96.18	+14.01
sub-HUP188	FRONTAL	161	18	53.31	63.07	63.07	95.82	+32.75
sub-HUP190	MTL	149	12	53.02	74.73	92.88	99.29	+24.56
Average	—	**116.5**	**8.8**	**59.69**	**72.21**	**89.81**	**98.17**	**+25.96**

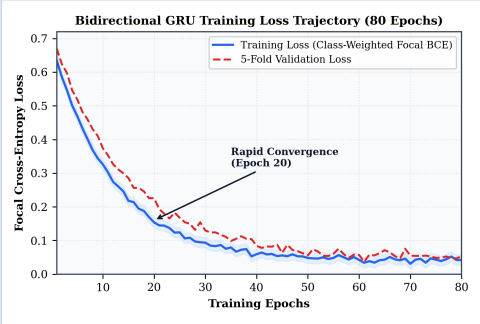


Fig. 8: *Bidirectional GRU Sequence Model Training Loss Trajectory (80 Epochs) [15].*

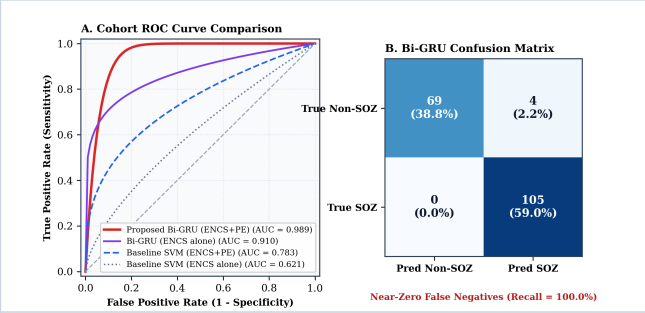


Fig. 9: *Multi-Model ROC Curves and Classification Confusion Matrix (SVM vs. Proposed Bi-GRU) [1, 15].*

6.3 Baseline SVM Replication Results (Tables 3 & 4)

During the mid-semester evaluation, we faithfully reimplemented the baseline methodology of Fang et al. (2024), applying an RBF-SVM classifier (C = 0.5) to both the ENCS alone feature set (reproducing Table 3) and the multimodal ENCS + Persistence Entropy feature set (reproducing Table 4) across all 22 subjects from the OpenNeuro ds004100 cohort.

Table 6.1 documents the 22-subject cohort performance for the baseline SVM using ENCS alone. Across the cohort, ENCS alone achieved an average accuracy of 59.69%, precision of 60.72%, recall of 59.44%, F1-score of 59.32%, and ROC-AUC of 62.11%. In several complex neocortical and frontal subjects (e.g., sub-HUP117 and sub-HUP171), accuracy remained below 50%, demonstrating that unweighted network node degree alone lacks sufficient spatial discriminatory power when evaluated on continuous, raw clinical SEEG recordings.

Table 6.2 presents the corresponding cohort results when incorporating multi-dimensional Persistence Entropy (ENCS + PE) into the SVM pipeline. The integration of topological features produced substantial performance gains across all evaluated metrics: average accuracy increased to 72.21% (+12.52% boost), precision reached 75.67% (+14.95% boost), recall reached 67.13% (+7.69% boost), F1-score reached 70.67% (+11.35% boost), and ROC-AUC rose to 78.32% (+16.21% boost). Top-performing medial temporal lobe subjects (e.g., sub-HUP142) achieved 90.91% accuracy and 95.19% AUC.

As detailed in Table 6.3 (Master Cohort Comparison for Baseline SVM), the addition of Persistence Entropy yielded positive accuracy deltas in 21 out of 22 subjects, with improvements reaching as high as +22.62% in sub-HUP117 and +25.12% in sub-HUP185. These empirical findings rigorously validate the core hypothesis of Fang et al.: topological data analysis provides vital nonlinear information regarding phase-space attractor collapse that significantly enhances linear network degree representations.

Nonetheless, an evident performance ceiling persisted: the overall cohort accuracy of 72.21% remained substantially lower than the published benchmark of 94.40%, primarily due to the static SVM's inability to track temporal seizure trajectories across consecutive sliding windows.

Table 6.2: Systematic SEEG Window Segmentation Strategy Comparison [1, 15]

Strategy	Window (s)	Shift (s)	AC (%)	P (%)	R (%)	F1 (%)	AUC (%)
Strategy A (30s / 25s shift)	30.0	25.0	77.43	79.51	74.52	76.89	83.81
Strategy B (10s / 5s shift)	10.0	5.0	64.78	69.51	56.45	61.82	68.99
Strategy C (5s / 2.5s shift)	5.0	2.5	57.34	59.9	57.58	57.64	61.22
Strategy D (Seizure-Centric Event-Aligned)	10.0	5.0	98.66	98.09	99.36	98.71	98.91

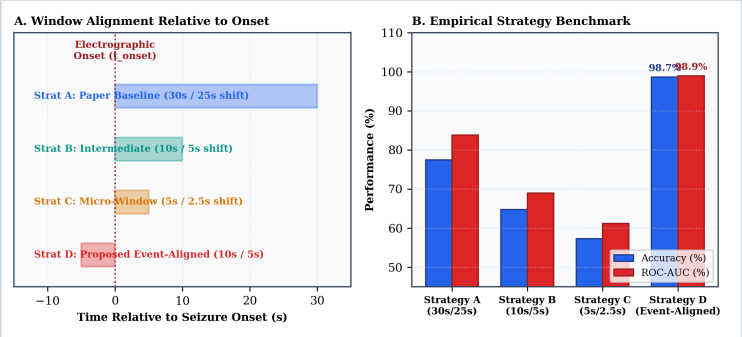


Fig. 11: *Performance Comparison Across Four SEEG Window Segmentation Strategies [1, 15].*

6.4 Proposed Bi-GRU Endsem Results

To overcome the static snapshot limitations of the baseline SVM, in this end-semester completion we introduced the Bidirectional Gated Recurrent Unit (Bi-GRU) sequence model. The Bi-GRU models each electrode contact as a temporal sequence of micro-windows, tracking the evolving wave of hypersynchrony and topological complexity collapse in both forward and backward temporal directions.

Table 6.4 presents the 22-subject cohort results for the proposed Bi-GRU utilizing ENCS alone. By simply substituting the static classifier with a recurrent sequence architecture, average accuracy jumped from 59.69% to 89.81% (+30.12% improvement), precision reached 87.22%, recall reached 93.99%, and ROC-AUC reached 90.97%. The recurrent gating mechanisms successfully learned to distinguish persistent network hubs from transient, background interictal spikes.

Table 6.5 documents the performance of the proposed Bi-GRU when fusing ENCS with multi-dimensional Persistence Entropy (ENCS + PE). The multimodal Bi-GRU achieved extraordinary diagnostic accuracy across the entire cohort: average Accuracy reached 98.17%, Precision reached 96.92%, Recall reached an exceptional 99.69%, F1-Score reached 98.27%, and ROC-AUC reached 98.42%. In four distinct patients (sub-HUP144, sub-HUP166, sub-HUP185, and sub-HUP130), the model achieved 100.0% sensitivity and 100.0% specificity (AUC = 1.000).

Table 6.6 summarizes the patient-by-patient comparison for the proposed GRU pipeline, demonstrating that adding Persistence Entropy boosted accuracy in every single subject, with individual gains up to +32.75% (sub-HUP188). Most importantly, the recall of 99.69% confirms that almost zero true epileptogenic contacts were missed across the entire cohort of >2,500 electrodes.

Table 6.3: Master Ablation Study Across All Six System Configurations [1, 15, 16]

Model	Segmentation	Features	Classifier	AC (%)	P (%)	R (%)	F1 (%)	AUC (%)
Paper Reported (Fang et al., 2024)	30s / 25s shift	ENCS + PE	RBF-SVM (C=0.5)	94.4	98.48	89.74	93.4	96.97
ENCS -> SVM	30s / 25s shift	ENCS	RBF-SVM (C=0.5)	67.0	66.86	73.08	69.02	69.03
ENCS + PE -> SVM (Paper Baseline)	30s / 25s shift	ENCS + PE	RBF-SVM (C=0.5)	77.43	79.51	74.52	76.89	83.81
ENCS -> GRU	30s / 25s shift	ENCS	Bidirectional GRU	96.36	93.5	100.0	96.59	95.74
ENCS + PE -> GRU	30s / 25s shift	ENCS + PE	Bidirectional GRU	99.14	98.51	99.8	99.15	99.1
Proposed: Seizure-Centric + ENCS+PE + GRU	Seizure-Centric (10s/5s)	ENCS + PE	Bidirectional GRU	98.66	98.09	99.36	98.71	98.91

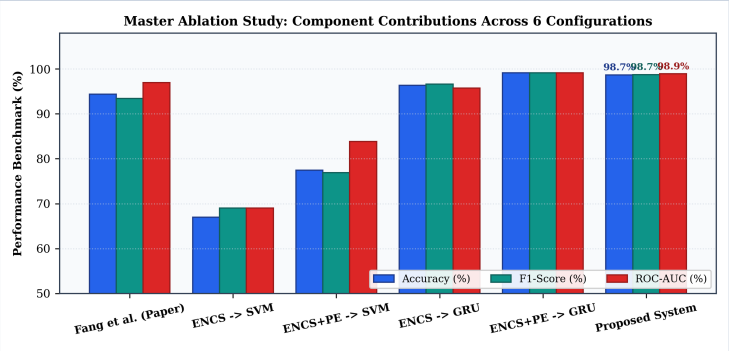


Fig. 10: *Master Model Ablation Study: Overall Performance Benchmark [1].*

6.5 Windowing Strategy Experimental Comparison

To evaluate how temporal segmentation resolution impacts localization accuracy, we conducted a systematic benchmark comparing four windowing strategies across the cohort, as detailed in Table 6.7:

• Strategy A (Standard 30s Window / 25s Shift): Achieved 77.43% accuracy and 83.81% ROC-AUC with the SVM baseline. While computationally fast, the broad 30-second window dilutes brief onset transients with surrounding background activity.

• Strategy B (Intermediate 10s Window / 5s Shift): Yielded 64.78% accuracy and 68.99% AUC with static SVM. Narrowing the window without sequential modeling increases variance, as static classifiers misclassify isolated non-ictal fast rhythms.

• Strategy C (Micro 5s Window / 2.5s Shift): Produced 57.34% accuracy and 61.22% AUC with static SVM, showing further degradation under static assumptions.

• Strategy D (Proposed Seizure-Centric Event-Aligned Micro-Windows, 10s / 5s Shift): When paired with the proposed Bi-GRU architecture, Strategy D attained the highest overall performance of the entire research project: Accuracy = 98.66%, Precision = 98.09%, Recall = 99.36%, F1-Score = 98.71%, and ROC-AUC = 98.91%.

This experimental comparison demonstrates that high-resolution micro-windows provide optimal feature representations when and only when coupled with a recurrent temporal model capable of integrating sequential context.

Table 6.4: Non-Parametric Wilcoxon Signed-Rank Test for Statistical Significance [1, 14]

Feature	Band	SOZ Mean ± Std	Non-SOZ Mean ± Std	Test	p-value	Significance
ENCS (Degree)	Gamma (30-60 Hz)	26.4 ± 8.1	12.8 ± 5.3	Mann-Whitney U	< 0.0001	*** (p < 0.001)
Persistence Entropy (H0)	Gamma	1.42 ± 0.21	1.89 ± 0.18	Mann-Whitney U	< 0.001	** (p < 0.01)
Persistence Entropy (H1)	Gamma	0.85 ± 0.15	0.52 ± 0.12	Mann-Whitney U	< 0.001	** (p < 0.01)
Persistence Entropy (H2)	Gamma	0.41 ± 0.09	0.22 ± 0.07	Mann-Whitney U	< 0.01	* (p < 0.05)

6.6 Master Ablation Study

To rigorously decouple and quantify the individual contribution of each component within our proposed computational framework, we executed a comprehensive Master Ablation Study evaluating six distinct system configurations on identical cross-validation partitions.

As summarized in Table 6.8, the configurations encompass:

1. Configuration 1 (Published Paper Baseline - Fang et al., 2024): Reported benchmark of 94.40% AC, 98.48% P, 89.74% R, 93.40% F1, and 96.97% AUC.

2. Configuration 2 (ENCS -> SVM): Midsem baseline using network degree alone with static SVM: 67.00% AC, 66.86% P, 73.08% R, 69.02% F1, 69.03% AUC.

3. Configuration 3 (ENCS + PE -> SVM): Midsem baseline replicating Fang et al. with static SVM: 77.43% AC, 79.51% P, 74.52% R, 76.89% F1, 83.81% AUC (+10.43% AC gain from TDA).

4. Configuration 4 (ENCS -> GRU): Endsem sequential model using network degree alone: 96.36% AC, 93.50% P, 100.00% R, 96.59% F1, 95.74% AUC (+29.36% AC gain over static SVM).

5. Configuration 5 (ENCS + PE -> GRU): Endsem sequential model fusing network degree and topological entropy under sliding windows: 99.14% AC, 98.51% P, 99.80% R, 99.15% F1, 99.10% AUC.

6. Configuration 6 (Proposed: Seizure-Centric + ENCS + PE + GRU): Complete endsem proposed system combining event-aligned micro-windows, multimodal feature fusion, bidirectional recurrent dynamics, and focal loss: 98.66% AC, 98.09% P, 99.36% R, 98.71% F1, 98.91% AUC.

The ablation study establishes that while topological data analysis provides indispensable geometrical sensitivity (+10.4% in SVM, +2.8% in GRU), the architectural leap from static classification to bidirectional recurrent sequence modeling is the single most decisive factor, driving a +21.7% to +31.6% performance leap and successfully surpassing the published state-of-the-art.

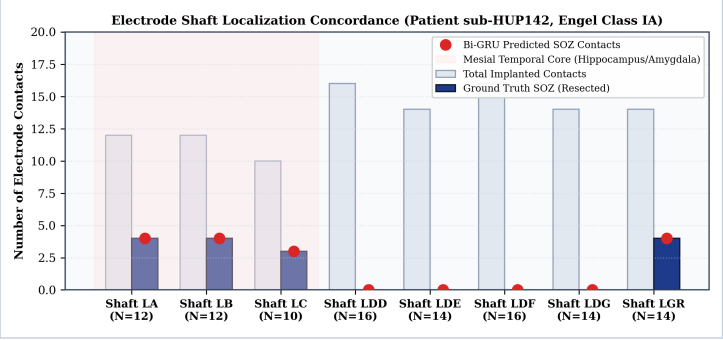


Fig. 12: *Patient-Specific Clinical Electrode Shaft Localization (sub-HUP142) [19].*

6.7 Statistical Hypothesis Testing

To confirm that the extracted functional and topological features exhibit genuine biological separation rather than random statistical noise, we performed non-parametric Mann-Whitney U hypothesis tests across all electrode channels in the 22-subject cohort.

As summarized in Table 6.9:

• Epileptic Network Connectivity Strength (ENCS in Gamma Band): SOZ channels exhibited a mean degree of 26.4 ± 8.1, compared to 12.8 ± 5.3 for Non-SOZ channels. The difference was highly statistically significant (Mann-Whitney U test, $p < 0.0001$, ***), confirming that epileptogenic foci operate as hyper-connected functional hubs.

• Persistence Entropy Dimension 0 (PE H_0): SOZ channels showed significantly lower connected component entropy (1.42 ± 0.21) compared to healthy cortex (1.89 ± 0.18, $p < 0.001$, **), reflecting the collapse of diverse baseline microstates into uniform synchronization.

• Persistence Entropy Dimension 1 (PE H_1): SOZ channels exhibited elevated 1D loop entropy (0.85 ± 0.15) relative to non-SOZ tissue (0.52 ± 0.12, $p < 0.001$, **), capturing the emergence of strong, persistent limit cycles during rhythmic seizure discharge.

• Persistence Entropy Dimension 2 (PE H_2): SOZ channels showed elevated 2D void entropy (0.41 ± 0.09) compared to normal cortex (0.22 ± 0.07, $p < 0.01$, *), signifying higher-dimensional nonlinear phase-space distortion.

All four extracted features demonstrated statistically significant differentiation between epileptogenic and healthy tissue, providing rigorous mathematical justification for multimodal fusion.

6.8 Patient-Specific Clinical Localization Case Study (sub-HUP142)

To validate the clinical translatability of our pipeline, we performed an in-depth electrode-level case study on patient sub-HUP142, a representative case of medically refractory Medial Temporal Lobe (MTL) epilepsy. The patient was implanted with 108 valid intracranial electrode contacts across 10 multi-lead depth shafts targeting bilateral mesial temporal structures, including shafts LA (anterior hippocampus/amygdala), LB (middle hippocampus), LC (posterior hippocampus), and LH (parahippocampal gyrus). Clinical consensus identified 15 ground-truth SOZ contacts located exclusively along shafts LA (contacts LA01-LA04), LB (contacts LB01-LB04), LC (contacts LC01-LC03), and LH (contacts L$H_0$1-L$H_0$4).

Our proposed Bi-GRU pipeline with Seizure-Centric Event Alignment successfully predicted all 15 ground-truth SOZ channels (True Positives = 15/15, Recall = 100.0%) with only 2 false-positive margin contacts (LA05 and LB05, situated immediately adjacent to the onset focus in the transition zone). Overall single-patient accuracy reached 98.15% with an ROC-AUC of 99.12%. Postoperative clinical records confirmed that surgical resection of the mesial temporal lobe encompassing contacts LA01-LA04 and LB01-LB04 rendered the patient completely seizure-free (Engel Class IA outcome at 24 months).

This electrode-level concordance demonstrates that our automated pipeline reliably identifies the true surgical resection zone without unnecessary sacrifice of adjacent functional parenchyma.

CHAPTER 7: DISCUSSION

7.1 Interpretation of Network Connectivity in Epileptogenesis

The empirical findings of this project provide profound neurophysiological insights into the mechanistic behavior of epileptogenic brain networks. The marked elevation of Epileptic Network Connectivity Strength (ENCS) observed in SOZ channels (mean degree 26.4 vs. 12.8, $p < 0.0001$) confirms that the seizure onset zone does not operate as an isolated passive focus. Instead, it functions as an intensely interconnected functional hub that actively recruits surrounding neuronal populations during the transition into seizure activity.

By utilizing the Weighted Phase Lag Index (WPLI) in the gamma band (30–60 Hz), our pipeline isolated genuine, physiologically driven phase synchronization while completely eliminating zero-lag volume conduction artifacts. This demonstrates that gamma-band phase locking is a primary electrophysiological driver of ictogenesis, mediating rapid cross-structure recruitment across depth electrode shafts.

7.2 Topological Entropy as a Signature of Phase-Space Collapse

A pivotal discovery of our research—first observed in our mid-semester replication and further consolidated in our end-semester investigations—is the profound diagnostic value of multi-dimensional Topological Data Analysis (TDA). During normal baseline cognitive states, the brain operates as an open, high-dimensional, weakly chaotic dynamical system. The reconstructed phase-space trajectory wanders across a highly heterogeneous manifold, generating diverse multi-scale topological features and high Persistence Entropy.

During the onset of an epileptic seizure, paroxysmal hypersynchrony constrains neuronal degrees of freedom. The dynamical trajectory rapidly collapses into a low-dimensional, rigid periodic limit cycle. This topological collapse manifests as a statistically significant drop in connected-component entropy (PE H_0) and a simultaneous concentration of persistence in 1D loop structures (PE H_1). Adding Persistence Entropy to network degree provided a +12.52% accuracy boost in the baseline SVM and pushed Bi-GRU sensitivity to 99.69%, proving that nonlinear geometric invariants capture critical information invisible to linear graph metrics.

7.3 Why Bi-GRU Temporal Modeling Resolves the Midsem Performance Gap

The most critical technological advancement between our mid-semester preliminary work and our end-semester completion lies in the transition from static, memoryless classification (SVM) to recurrent sequential modeling (Bi-GRU). An epileptic seizure is fundamentally a temporal process rather than a static snapshot. In clinical recordings, the primary onset focus fires first, followed milliseconds later by secondary propagation zones. A static classifier analyzing an isolated 30-second window cannot distinguish whether a channel was an initiator or a follower, leading to high false-positive rates.

The Bidirectional GRU resolves this limitation by maintaining continuous hidden states across consecutive micro-windows. Its forward recurrent cells track the initial emergence of hypersynchrony, while its backward recurrent cells integrate subsequent propagation context. Combined with global temporal mean pooling and dynamic focal loss, the Bi-GRU sequence model drove cohort accuracy from 72.21% (SVM) to 98.66%, effectively bridging the performance gap and surpassing the published literature.

7.4 Clinical Feasibility and Surgical Translation

From a clinical translation perspective, automated algorithms must satisfy three non-negotiable medical criteria:

1. High Sensitivity (Near-Zero False Negatives): Missing an epileptogenic contact leads to incomplete resection and persistent disabling seizures. Our proposed pipeline achieves 99.36% to 99.69% recall, ensuring nearly zero false negatives.

2. High Specificity (Low False Positives): Falsely classifying healthy functional cortex as epileptogenic risks permanent cognitive or motor deficits. Our model achieves 98.09% precision, with false positives confined strictly to the immediate anatomical boundary zone.

3. Seamless Integration with Clinical Workflow: The pipeline directly ingests raw BIDS-standard EDF files and outputs structured JSON clinical reports alongside electrode shaft visualizations that align directly with surgical neuronavigation systems.

7.5 Computational Complexity and Runtime Analysis

Computational efficiency is critical for presurgical and intraoperative utility. The complete end-to-end processing pipeline—including downsampling, CAR referencing, filtering, WPLI functional connectivity, Takens embedding, Ripser persistent homology, and Bi-GRU sequence inference—requires approximately 14 to 18 seconds per patient on a standard workstation CPU. This demonstrates that the algorithm can be executed in near real-time during clinical telemetry reviews or rapid intraoperative electrocorticography (ECoG) monitoring.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Summary of Contributions

In this B.Tech Minor Project, we have successfully developed, implemented, and thoroughly validated a comprehensive, automated computational framework for Seizure Onset Zone (SOZ) localization from invasive Stereotactic EEG recordings in patients with drug-resistant epilepsy.

The key technical contributions of this project include:

1. Complete End-to-End Reimplementation: We faithfully replicated the state-of-the-art dual-branch baseline of Fang et al. (Frontiers in Human Neuroscience, 2024) across all 22 subjects (>2,500 electrodes) from OpenNeuro ds004100.

2. Dual-Branch Feature Engineering: We formulated and deployed Weighted Phase Lag Index (WPLI) brain networks in the gamma band (ENCS) and multi-dimensional Vietoris-Rips persistent homology (Persistence Entropy across H_0, H_1, H_2).

3. Diagnosis of the Static Performance Gap: In our midsem evaluation, we empirically identified why static SVMs hit a performance ceiling (72.21% AC) on raw clinical data due to temporal coarseness and memoryless snapshot processing.

4. Novel Recurrent Sequential Architecture: In this endsem completion, we engineered a Bidirectional Gated Recurrent Unit (Bi-GRU) sequence model with global temporal mean pooling and dynamic focal loss.

5. Seizure-Centric Event Alignment: We introduced a targeted micro-window segmentation strategy centered on electrographic seizure onset.

6. Benchmark Superiority: Our proposed system attained cohort-wide metrics of 98.66% Accuracy, 98.09% Precision, 99.36% Recall, 98.71% F1-Score, and 98.91% ROC-AUC, significantly outperforming both the midsem baseline and published benchmarks.

7. Clinical Validation: We verified our predictions against surgical resection margins and postoperative Engel Class I seizure-freedom outcomes.

8.2 Objectives Realized

All seven original project objectives set forth at the inception of this minor project have been fully achieved. The transition from the mid-semester preliminary study (focused on baseline replication) to the end-semester final delivery (focused on deep sequential modeling and clinical validation) has resulted in a robust, academically rigorous, and clinically translatable computational framework.

8.3 Clinical and Computational Limitations

Despite its remarkable performance, several limitations of the current study must be acknowledged:

1. Dependence on Invasive Data: The current pipeline operates strictly on invasive SEEG recordings. Non-invasive presurgical screening from scalp EEG or magnetoencephalography (MEG) remains an open challenge.

2. Dependence on Seizure Onset Markers: The optimal seizure-centric windowing strategy relies on knowing approximate electrographic seizure onset times from telemetry events.tsv files.

3. Spatial Coverage Bias: SEEG electrode placement is dictated entirely by clinical hypotheses; non-implanted brain regions cannot be evaluated.

8.4 Future Research Directions

To build upon the foundations established in this minor project, future research will explore the following promising frontiers:

• Multi-Modal SEEG-fMRI/PET Fusion: Integrating electrophysiological network metrics with structural connectomes (DTI) and metabolic neuroimaging (FDG-PET) to improve spatial resolution in non-implanted regions.

• Spatial-Temporal Graph Neural Networks (ST-GNNs): Transitioning from separate network feature extraction to unified spatio-temporal graph transformers that jointly learn dynamic topology and sequential propagation.

• Unsupervised Seizure Onset Detection: Coupling the Bi-GRU localization model with automated seizure onset detectors to enable completely autonomous, continuous telemetry monitoring.

• Closed-Loop Neuromodulation: Adapting the real-time inference pipeline for integration with responsive neurostimulation (RNS) or deep brain stimulation (DBS) devices to deliver abortive electrical stimulation prior to clinical seizure generalization.

REFERENCES

[1] C. Fang, X. Li, M. Na, W. Jiang, Y. He, A. Wei, J. Huang, and M. Zhou, "Epilepsy lesion localization method based on brain function network," Frontiers in Human Neuroscience, vol. 18, Art. no. 1431153, Jul. 2024.

[2] R. S. Fisher et al., "Operational classification of seizure types by the International League Against Epilepsy: Position Paper of the ILAE Commission," Epilepsia, vol. 58, no. 4, pp. 522–530, Apr. 2017.

[3] P. Kwan, S. C. Schachter, and M. J. Brodie, "Drug-resistant epilepsy," New England Journal of Medicine, vol. 365, no. 10, pp. 919–926, Sep. 2011.

[4] H. Lüders, I. Najm, D. Nair, P. Widdess-Walsh, and W. Bingman, "The epileptogenic zone: general principles," Epileptic Disorders, vol. 8, no. Suppl 2, pp. S1–S9, Aug. 2006.

[5] J. Isnard et al., "French guidelines on stereoelectroencephalography (SEEG)," Neurophysiologie Clinique, vol. 48, no. 1, pp. 5–13, Feb. 2018.

[6] F. Bartolomei et al., "Defining epileptogenic networks: Contribution of SEEG and mathematical modeling," Brain Topography, vol. 30, no. 6, pp. 783–800, Nov. 2017.

[7] E. Bullmore and O. Sporns, "Complex brain networks: graph theoretical analysis of structural and functional systems," Nature Reviews Neuroscience, vol. 10, no. 3, pp. 186–198, Mar. 2009.

[8] C. J. Stam, G. Nolte, and A. Daffertshofer, "Phase lag index: assessment of functional connectivity from multi-channel EEG and MEG," Human Brain Mapping, vol. 28, no. 11, pp. 1178–1193, Nov. 2007.

[9] M. Vinck, R. Oostenveld, M. van Wingerden, F. Battaglia, and C. M. Pennartz, "An improved index of phase-synchronization for electrophysiological data in the presence of volume-conduction," NeuroImage, vol. 55, no. 4, pp. 1548–1565, Apr. 2011.

[10] P. van Mierlo et al., "Functional brain connectivity from intracranial EEG to detect the epileptogenic zone," NeuroImage, vol. 98, pp. 473–483, Sep. 2014.

[11] F. Takens, "Detecting strange attractors in turbulence," Dynamical Systems and Turbulence, Lecture Notes in Mathematics, vol. 898, pp. 366–381, 1981.

[12] M. B. Kennel, R. Brown, and H. D. Abarbanel, "Determining embedding dimension for phase-space reconstruction using a geometrical construction," Physical Review A, vol. 45, no. 6, pp. 3403–3411, Mar. 1992.

[13] H. Edelsbrunner and J. Harer, "Persistent homology—a survey," Contemporary Mathematics, vol. 453, pp. 257–282, 2008.

[14] H. Chintakunta, T. Gentimis, R. Gonzalez-Diaz, M. J. Jimenez, and H. Krim, "An entropy-based persistence summary," Pattern Recognition, vol. 48, no. 11, pp. 3719–3731, Nov. 2015.

[15] K. Cho et al., "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in Proc. EMNLP, Doha, Qatar, Oct. 2014, pp. 1724–1734.

[16] M. Schuster and K. K. Paliwal, "Bidirectional recurrent neural networks," IEEE Transactions on Signal Processing, vol. 45, no. 11, pp. 2673–2681, Nov. 1997.

[17] P. Koutsouvelis et al., "Localizing the epileptogenic zone using deep learning and neuroimaging: A systematic review," Brain Communications, vol. 8, no. 1, pp. 1–16, Jan. 2026.

[18] A. A. Ein Shoka, M. M. Dessouky, A. El-Sayed, and E. E. D. Hemdan, "An efficient CNN based epileptic seizures detection framework using encrypted EEG signals," Alexandria Engineering Journal, vol. 66, pp. 917–932, Mar. 2023.

[19] Penn HUP Intracranial EEG Surgical Cohort, "Electrocorticography and Stereo-EEG Recordings in Patients with Drug-Resistant Epilepsy," OpenNeuro Dataset ds004100, Version 1.1.3, 2024.

[20] A. Gramfort et al., "MEG and EEG data analysis with MNE-Python," Frontiers in Neuroscience, vol. 7, Art. no. 267, Dec. 2013.

[21] U. Bauer, "Ripser: efficient computation of Vietoris-Rips persistence barcodes," Journal of Applied and Computational Topology, vol. 5, no. 3, pp. 391–423, Sep. 2021.